



VIKAS GOEL

Machine Learning/Vision and Data Science Expert

CONTACT

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Date of Birth: 3rd April, 1972 (03 – 04 – 1972)

Native Place: Muzaffar Nagar (UP) INDIA

Current Location: Mohali (Chandigarh) INDIA

PROFILE SUMMARY

Have strong 20 years of experience in Digital Signal/Image Processing, Data Science Analysis using Matlab/Simulink and Python along with embedded system design and development using different microcontrollers using C/C++.

ACHIEVEMENT

An Indian Patent (No. 446/Del/2001) on **“Modeling of Automobile Filter Media”** (Software Patent).

KEY SKILLS

Matlab, Simulink, Python-3, C/C++ , R-Studio

TECHNOLOGY WORKED ON:

NLP, Regression, Clustering, Classification, AIML, ANN, SVM, Curve Fitting, ANOVA, K-Means, KNN, Hierarchical clustering, PCA, Object Tracking, Machine Learning (ML), AI

LIBRARIES WORKED ON:

Matplotlib, NumPy, PIL, Pandas, Scipy, Sklearn, FFT, NLP, Histogram Processing, tkinter, ARIMA, SARIMA, PCA, Interpolation, OpenCV

CERTIFICATES

CutShort Certified Python – Basic by CutShort on June 23, 2020

Quality System NABL - 101 for Laboratories (07/1999 – 07/1999)

Mission-10X Workshop for Effective Teaching Skills (06/2011 – 06/2011)

PROFESSIONAL QUALIFICATION

M.Tech. (Instrumentation) in 1998

From NIT Kurukshetra (Haryana) INDIA

PROFESSIONAL/WORK EXPERIENCE

Present: (Since June 2020)

Professor – Machine Learning and Data Science

at Chandigarh Group of Colleges, Landran (Mohali) Punjab INDIA

Engaged in Post-Graduate teaching: Machine Learning using Python, DSP and Image Processing using Matlab/Simulink and Embedded System using C/C++

Experience (Past)

❖ **Sr. Manager –Digital Signal/Image Processing (DSP) and Data Science**
at Labat Asia Pvt. Ltd., Mohali (Punjab) India from Jan. 20218 to June 2020

❖ **Sr. Manager – Data Science and Image processing – 7 Years**
at Centre for Dev. of Advanced Computing (C-DAC), Mohali (Punjab) India from Aug. 2011 to Dec. 2017

❖ **Sr. Manager – DSP and Image Processing - 2 Years**
at Recorders & Medicare Systems Pvt. Ltd., Chandigarh India from Jun. 2007 to Oct. 2009

❖ **Project Leader – Model Based Development (MBD) - 4 Years**
at National Res. & Tech. Consortium (NRTC) Parwanoo HP India from Jun. 1998 to 2009

Executed projects for:

- Monte Carlo Theory of Random Numbers for Filter Media Simulation using C/C++ for Purolator India Ltd., Parwanoo.
- **Engine Oil Soot Generation/Simulation using C/C++ Project for Indian Oil Corporation, Faridabad. (Patented Project)**

Projects Executed (acomplished) in the capacity of TeamLeader:

S. No.	Project Title	Capacity/Role	Budget	Funding Agency
1	Tone Generation and Analysis using MATLAB for Audiometer Development (Audio Signal Processing)	PI	5.0 Lacs	LAPL, Mohali
2	Crop Analysis using Image Processing in MATLAB	Co-PI	92.00Lacs	DeitY, New Delhi
4	Simulation of Automobile Filter Media using Image Processing in MATLAB (Patented)	Team Leader	10.00 Lacs	PIL, Parwanoo
5	Modelling of Engine Oil Thickening due to Soot Generation using MATLAB (Clustering and Segmentation algorithms)	Team Leader	5.00 Lacs	IOC, Faridabad

Organization	Designation	Nature of Duties	Period
CDAC, Mohali	Sr. Proj. Manager – R&D	R&D and M.Tech. Teaching (Digital Signal and Image Processing)	Aug. 2011 – Dec. 2018
IET, Bhaddal	HoD – EIE	Deptt. Head	Nov. 2009 – August 2011
Rec. & Med. Systems Pvt. Ltd., Phase – 1, Ind. Area, Chandigarh	Asst. Manager – R&D	Worked on development of Automatic Finger Print Identification	June 2007 to Oct. 2009
Clarity Medical Pvt. Ltd., C – 84, Ind. Area, Phase – 7, S.A.S. Nagar, Mohali - 160 059 (Pb.)	Asst. Manager - R&D	Worked on LPC 2138 micro controller based hardware and software development for medical diagnostic instruments	Sept. 2006 to May 2007
Micro Mation Pvt. Ltd., E-38, Phase VIII, Ind. Area, S.A.S. Nagar, Mohali - 160 059 (Pb.)	Assistant Manager - R&D	Worked on a project for productionize the “Clinical Chemistry Analyser” based on <i>electro-optical instrumentation</i> using <i>embedded micro-controller (8031)</i>	Sept. 2002 to August 2006
National Res. & Tech. Co., # 15, Sector-3, Parwanoo - 173 220 (HP)	Scientific Officer	Development of S/w for design and development of Automobile Filters, Engine Oil and Image Processing based Instrumentation -	June 1998 to Sept. 2002 (Four Years)
Allied Nippon Ltd. A-12, Site IV, Ind. Area, Sahibabad - 201 010(UP)	Assistant Engineer – Instrumentation	Calibration + Monitoring and Maintenance of Process Control Instruments like temp. Controllers, pressure gauges, pH meters, conductivity meters.	August 1995 to Sept. 1996 (One Year)

PAST EXPERIENCE:

Research Publications:

- Goel V., Sohi B.S. and Singh H., **“Statistical Optimization of Radial Vectors for Pattern Reconstruction/Retrieval,”** in 3rd International Conference on Computational Intelligence and Communication Technology-2017, ABES College, Ghaziabad (India)
- “Radial Vectored Features for Pattern Enveloping and Retrieval”**, Vikas Goel, B.S. Sohi and Harpal Singh, Journal of Optoelectronics And Advanced Materials, Vol. 18, No. 9-10, September - October 2016, p. 878 – 883
- “Interactive Intelligent Shopping Cart Using RFID and Zigbee Module”**, Vikas Goel and Shalu Dhaut, 5th International Conf. on Advances in Computing, Communications and Informatics, 21-24 Sept. 2016, Jaipur, Rajasthan, INDIA
- “Mathematical Modeling of Engine Oil Thickening due to Soot Generation”**, **Sujit K. Sinha and Vikas Goel**, International Symposium on Fuels and Lubricants, New Delhi, 7-9 Oct., 2002
- “Intelligent Mining System (IMS) as a New Mine Model for Enhanced Safety and Efficiency”**, Sujit K. Sinha, Sudhir Kumar and Vikas K. Goel, Indian Conference on Computer Applications in Mining Industries (ICCAMI), 29-31 March, 2001, New Delhi
- “Image Processing Based Visual Analysis System for Material’s Surface”**, S. Sarkar, Vikas Goel and Kiran Gupta, National Conference on Recent Developments in Disordered Materials. 15-16 March 2001, Panjab University, Chandigarh
- “AUTOMATIC VISUAL INSPECTION SYSTEM FOR SURFACE DEFECTS USING IMAGE PROCESSING”**, VIKAS GOEL AND H.K. SARDANA, CSIO, CHANDIGARH-160 020 INDIA, NATIONAL SEMINAR ON ADVANCE INSTRUMENTATION, 29–31 JAN., 2001,
- “A Precision Thermocouple Calibration System”**, Sujit k, Sinha, Sanjeev Tyagi, Vikas Goel, Vishal Gupta and N.S. Bhalla, National Seminar on Advance Instrumentation (NSAI), 29–31 Jan., 2001, CSIO, Chandigarh

Quality Training Attended:

S. No.	Training Course	Place	Duration
1	FDP on VLSI	C-DAC, Mohali	27-29 Nov. 2017
2.	Good Lab Practices	IET, Bhaddal	01-06-2011
3.	Mission-10X Workshop for Effective Teaching Skills	IET, Bhaddal	13 – 17 June 2011
4.	Quality System NABL - 101 for Laboratories	Indian Institute of Quality Management, Jaipur	Three Days 26 – 28 July 1999

5.	WINISIS Library Documentation System	Himachal Institute of Public Administration, Shimla	Five Days 6-10 Nov., 2000
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Expert Lecture/Training/Workshop Session Conducted:

S. No.	Training Course	Place	Date
1	Digital Signal Processing	Krishna Eng. College, Mohan Nagar, Ghaziabad (UP)	13-06-2017
2	Digital Image Processing	JMIT, Radour, (Haryana)	20-02-2016
3	PID Control Systems and Signal Processing	RIET, Kharar	22-07-2014
4	Digital Image Processing	RMIT, MandiGobindgargh	07-12-2013

CERTIFICATES:

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y Society of India, NPL, New Delhi
- Life Member of Instrument Society of India, Bangalore
- Life Member of Institute of Science and Technical Education (ISTE), India

Date: 15-06-2020
Place: Mohali

(Vikas Kumar Goel)